

Topic 9

Probability

**Concept guide · Worked examples · Practice · Investigation and modelling
· Solutions**

Learning focus

Content explanations, original Cambridge-style practice and an investigation task tailored to International Mathematics 0607.

Topic map

9.1	Probability scale and complements
9.2	Relative frequency and sample spaces
9.3	Combined independent events
9.4	Tree diagrams and dependent events
9.5	Venn diagrams and conditional reasoning
9.6	Probability simulation

Core foundations and Extended applications are integrated. Practise exact methods, calculator work and the written reasoning required in investigations.

1. Single events

A probability lies between 0 and 1. For equally likely outcomes, $P(A) = \text{favourable outcomes} / \text{total outcomes}$. Complement $P(\text{not } A) = 1 - P(A)$. Experimental relative frequency approximates probability when trials are numerous, but is not guaranteed to match exactly.

Worked example

A fair die gives $P(\text{even}) = 3/6 = 1/2$; therefore $P(\text{odd}) = 1/2$.

Exam check: Check that the probabilities of mutually exclusive exhaustive outcomes add to 1.

2. Sample space and combinations

List the full sample space systematically using a table or tree. For either event A or event B, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. If events cannot occur together, the overlap is zero. With equally likely pair outcomes, count ordered pairs rather than just distinct totals.

Worked example

Two fair dice have 36 ordered outcomes. Sum 7 occurs in six pairs, so $P(\text{sum } 7) = 6/36 = 1/6$.

Exam check: (2,5) and (5,2) count separately when dice are distinguishable.

3. Independent events

For independent events, knowing one does not change the probability of the other. Multiply along tree branches and add probabilities of alternative paths. Replacement leaves the original probabilities unchanged; without replacement they change.

Worked example

A fair coin is tossed twice: $P(\text{two heads}) = 1/2 \times 1/2 = 1/4$; $P(\text{exactly one head}) = 1/4 + 1/4 = 1/2$.

Exam check: Check whether the item is replaced before drawing the second branch.

4. Dependent events and conditional information

Without replacement, both numerator and denominator may change after the first selection. $P(A \text{ given } B)$ uses the reduced group satisfying B as its denominator. A Venn diagram can locate that group: $P(A|B) = n(A \cap B) / n(B)$ for equally likely people.

Worked example

Of 12 pupils, 5 play chess, 4 music and 2 both. Among music pupils, $P(\text{chess} | \text{music}) = 2/4 = 1/2$.

Exam check: Do not divide by the full class when the wording says "given that" or "among those".

5. Reading trees and checks

The two outcomes at a branch sum to 1 at each split. Complete a tree before multiplying paths. If an event has several separate paths, add their path probabilities. As a final check, all terminal path probabilities must sum to 1.

Worked example

A bag contains 3 red and 2 blue counters; without replacement $P(\text{two red}) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$.

Exam check: A second draw probability may use four remaining counters, not five.

6. Probability simulation

A simulation imitates the stated probabilities and replacement conditions. Repeat many trials, record the relative frequency and compare with the theoretical probability. A difference in a small sample is expected, but a consistent large discrepancy suggests a faulty model or method.

Worked example

Two independent red draws from a 3-red/2-blue bag occur with probability $\frac{9}{25}=0.36$.

Exam check: Simulation outcomes need not equal the theoretical prediction exactly.

0607 · Technology and communication

Graphical calculator and Paper 6

Use simulation to estimate relative frequencies and compare them with a theoretical model; separate random variation from bias.

State the rule or model, display enough working to reproduce your result, test it against the given values and discuss at least one limitation.

Practice paper

Topic 9 · Probability

Instructions

Original practice, informed by June 2026 components. Section A: short multiple choice for revision without a calculator. Section B: show full written working. Section C: investigation and modelling with reasons and checks.

Candidate name:	
Class / date:	

Total: 40 marks

Section A · Multiple choice

Choose one answer for each question.

1. $P(A) = 0.35$. What is $P(\text{not } A)$? [1]

- A 0.35
- B 0.65
- C 1.35
- D 0.75

2. $P(\text{heads twice})$ for a fair coin? [1]

- A $1/2$
- B $1/4$
- C $1/3$
- D $1/8$

3. Probability of a 6 on a fair die? [1]

- A $1/3$
- B $1/6$
- C $1/4$
- D 6

4. A certain event has probability [1]

- A 0
- B 0.5
- C 1
- D 6

5. For independent events, $P(A \text{ and } B)$ equals [1]

- A $P(A)+P(B)$
- B $P(A)P(B)$
- C $P(A)/P(B)$
- D $1-P(A)$

6. A bag has 2 red and 3 blue. $P(\text{red})$ is [1]

- A $2/3$
- B $3/5$
- C $2/5$
- D $1/5$

7. With 3 red and 2 blue, $P(\text{red then red})$ without replacement? [1]

- A $9/25$
- B $3/10$
- C $1/5$
- D $2/5$

8. If $P(A)=0.5$, $P(B)=0.4$ and $P(A \cap B)=0.1$, find $P(A \cup B)$. [1]

- A 0.1
- B 0.8
- C 0.9
- D 1.0

Section B · Structured questions

9. A fair coin is tossed and a fair six-sided die is rolled.

(a) Find $P(\text{heads and six})$. [2]

(b) Find $P(\text{tails and an even number})$. [2]

(c) Find $P(\text{heads or tails})$. [1]

[Total: 5]

10. A bag contains 4 red and 3 blue counters. Two are selected without replacement.

(a) Find $P(\text{two red})$. [2]

(b) Find $P(\text{red then blue})$. [2]

(c) Find $P(\text{one of each colour})$.

[3]

[Total: 7]

11. Of 30 pupils, 16 take French, 13 take Spanish and 5 take both.

(a) Find how many take neither.

[2]

(b) Find $P(\text{French or Spanish})$.

[2]

(c) Find $P(\text{French given Spanish})$.

[2]

[Total: 6]

12. A biased spinner lands on A with probability 0.30, B with 0.45 and C otherwise. It is spun twice independently.

(a) Find $P(C)$.

[1]

(b) Find $P(\text{two C outcomes})$.

[2]

(c) Find $P(\text{exactly one A})$.

[3]

[Total: 6]

Section C · Investigation and modelling

Show the rule, test it with values and explain the result in context.

13. A bag contains 3 red and 2 blue counters; a counter is drawn, returned and drawn again.

(a) Find the theoretical probability of two reds. [2]

(b) Estimate how many two-red outcomes to expect in 100 trials. [2]

(c) Explain whether exactly that many must occur. [2]

(d) State how a simulation could test the model. [2]

[Total: 8]

Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Method, model, communication and verification matter. A numerical answer should follow from clear working.

Section A · Answers and reasons

1. B — 0.65. Use the complement: $1 - 0.35 = 0.65$.
2. B — $1/4$. Two independent heads probabilities multiply: $(1/2)^2$.
3. B — $1/6$. One favourable face among six equally likely faces.
4. C — 1. A certain event occurs on every outcome.
5. B — $P(A)P(B)$. Independent events multiply for the intersection.
6. C — $2/5$. Two of five counters are red.
7. B — $3/10$. Without replacement: $3/5 \times 2/4 = 3/10$.
8. B — 0.8. Use inclusion-exclusion: $0.5 + 0.4 - 0.1 = 0.8$.

Section B · Worked solutions

9. a) $(1/2)(1/6) = 1/12$ [2]. b) $(1/2)(3/6) = 1/4$ [2]. c) A coin always shows heads or tails, so 1 [1].
10. a) $(4/7)(3/6) = 2/7$ [2]. b) $(4/7)(3/6) = 2/7$ [2]. c) red-blue or blue-red = $(4/7)(3/6) + (3/7)(4/6) = 4/7$ [3].
11. a) Union = $16 + 13 - 5 = 24$, neither = 6 [2]. b) $24/30 = 4/5$ [2]. c) $5/13 = 5/13$ [2].
12. a) $1 - 0.30 - 0.45 = 0.25$ [1]. b) $0.25^2 = 0.0625$ [2]. c) A then not A or not A then A: $2(0.3)(0.7) = 0.42$ [3].

Section C · Investigation and modelling solutions

13(a). $(3/5)^2 = 9/25 = 0.36$ [2].

13(b). $0.36 \times 100 = 36$ [2].

13(c). No; individual trials are random and the observed count varies [2].

13(d). Simulate two independent draws with probabilities $3/5$ and $2/5$; repeat many times and compare relative frequency with 0.36 [2].

Self-check

Explain why your model fits the starting data, show an independent check and name a condition under which your prediction would be unreliable.

Reference: Cambridge IGCSE International Mathematics 0607 syllabus 2025-2027, <https://www.cambridgeinternational.org/Images/662472-2025-2027-syllabus.pdf>. Assessment conventions informed by the June 2026 papers and mark schemes supplied by the user. All practice questions are newly written.